

SRIHITA RAMANAN

Senior Software Engineer

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github.com/srihitaramanan | srihitaramanan.github.io | [US Patent Application US20240013318](https://www.uspto.gov/patent/applications/US20240013318)

SUMMARY

Senior Full-Stack Software Engineer with 6+ years building member-facing and internal-tooling experiences in fintech, payments, and risk. Leads end-to-end delivery across React/TypeScript frontends and backend services on AWS, with system design for performance, reliability, and observability at scale. Hands-on with AI/ML integration including RAG and LLM-powered workflows applied to support and conversational experiences. Inventor on a pending machine learning patent.

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript (ES6+), TypeScript, C#

Backend & Distributed Systems: Spring Boot, Spring MVC, Hibernate, Microservices, REST APIs, GraphQL, gRPC, Event-Driven Architecture, Apache Kafka, Guice

Databases: MySQL, Oracle, PostgreSQL, MongoDB, Redis, pgvector

AI & ML: RAG (Retrieval-Augmented Generation), Spring AI, vector search, ML model integration, LLM-assisted development, AI-driven log analysis, prompt engineering

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, CloudWatch, Auto Scaling, VPC, CloudFront, Route 53), Docker, Kubernetes, Terraform, Ansible, Chef, Jenkins, GitHub Actions, CI/CD

Frontend: React, Angular, Vue.js, HTML5, CSS3

Testing & Observability: JUnit, Mockito, Cucumber, Postman, Prometheus, Grafana, ELK Stack, Distributed Tracing

Practices: System Design, API Design, Agile/Scrum, Code Reviews, Performance Optimization

EXPERIENCE

eBay Inc.

Jan 2021 - Present

Senior Software Engineer | San Jose, CA

eBay Live Platform May 2025 - Present

- **Drove new GMV streams by launching the Live Platform across 4 international markets** (Canada, Australia, France, Italy) with minimal production rollout risk.
- **Cut non-compliant seller activity and strengthened buyer trust** by building backend services for seller onboarding, compliance enforcement, and multi-user access control.
- Integrated LLM-powered workflows into production to interpret unstructured inputs and complex policy edge cases, improving decision quality and reducing manual handoffs in support-adjacent flows.
- Optimized high-traffic streaming APIs with caching and response prioritization, and reduced incident resolution time via end-to-end observability and anomaly detection.

eBay Risk Platform May 2022 - Apr 2025

- **Enabled \$10M+ in incremental company GMV** by building scalable APIs and data pipelines for the Unified User Privilege Program, unlocking trusted-user access and faster customer refunds at scale.
- **Improved risk-decision accuracy and reduced manual exception handling** by designing risk evaluation logic combining internal risk signals with external identity systems and LLM-based interpretation of unstructured inputs.
- Led the Unified User Privilege Program end-to-end, architecting modular backend services with Guice for multi-signal risk evaluation, and migrated core services to Spring Boot for improved performance and reliability.
- Strengthened observability with automated dashboards, alerts, and LLM-assisted log summarization for faster incident triage.

eBay Payments Engineering Jan 2021 - May 2022

- **Supported mission-critical payments and compliance workflows at scale** by building high-performance backend services for eBay's tax service pipeline in Java, Spring, and Hibernate.
- **Improved tax-determination throughput and reliability** by architecting a flexible ML results-aggregation component serving model-driven decisions with dual read/write across data stores.
- Co-designed and trained a Tax Category Prediction ML model spanning data preparation and experimentation; basis for a pending US patent.
- Delivered an internal nuData caching and API platform end-to-end, including backend REST services, keyspaces, and a React/TypeScript self-service console used by other engineering teams; deployed on Docker and Kubernetes.

MaxVal IP Services

Jul 2017 - Jun 2018

Software Engineer | Coimbatore, India

- **Enabled secure, highly scalable application infrastructure** by designing and operating AWS cloud systems (EC2, Auto Scaling, VPC, RDS, CloudWatch, CloudFront, Route 53).
- **Ensured consistent, repeatable deployments across environments** by automating provisioning with Chef and Docker and owning CI/CD pipelines in Jenkins.
- Deployed and monitored containerized applications on Docker and Kubernetes, resolving runtime and resource issues.

PROJECTS

DocuMentor — RAG-Powered Document Q&A API (Work in Progress) github.com/srihitaramanan | Java, Spring Boot 3, PostgreSQL, pgvector

- Building a production-grade REST API for document Q&A using Retrieval-Augmented Generation (RAG), returning LLM-generated answers grounded in uploaded PDFs with inline citations.
- Designed the retrieval layer on PostgreSQL with pgvector for semantic vector search, and an ingestion pipeline that chunks documents and generates embeddings via Spring AI.

PATENTS

Machine Learning-Based Tax Category Prediction System [US Patent Application US20240013318](https://www.uspto.gov/patent/applications/US20240013318) (Pending)

- Inventor on a pending patent for an ML system automating tax category prediction, covering model design and a results-aggregation architecture for model-driven decisions.

EDUCATION

University of Houston-Clear Lake — M.S. Computer Science

Houston, TX | May 2020

Anna University — B.E. Computer Science

Chennai, India | May 2017